

Get Your Guide

A Localization Case Study

While browsing the GetYourGuide website in German, I observed sub-optimal machine translations. What primarily caught my eye were the activity titles that were consistently mis-localized in the target language **German** (de-DE) from source **English** (en-US).

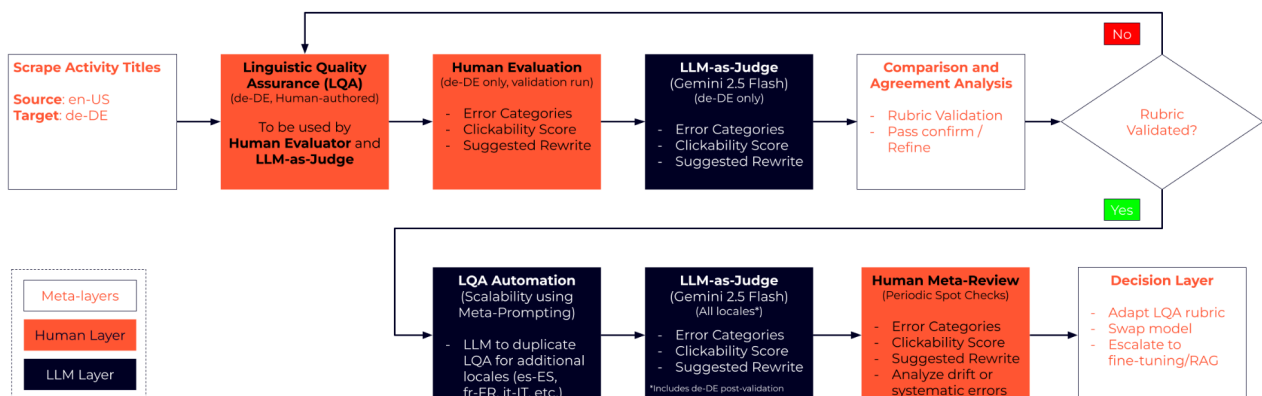
They contain multiple quality issues, including unnatural *Fluency*, *Terminology* misalignment, lack of *Accuracy* for proper nouns, *Grammatical* errors, and *Tone* mismatch related to product appeal.

A prime example of the poor localization on the website is the Hero Banner itself, whose German translation **Erlebnisse, für die es sich zu reisen lohnt** from the original English **Experiences worth traveling for** is translated word-by-word and comes across as rather rigid and forced for German audience. It contains the so-called *Translationese* issue affecting the *Fluency* / *Style* expected from a high quality translation. A more natural Hero Banner could read: **Erlebnisse, die eine Reise wert sind** or **Unvergessliche Erlebnisse, für die sich jede Reise lohnt**, among others. The problematic part in the original translation is the use of **zu reisen**, which creates an abrupt break in the sentence flow.

This case study aims to identify scalable approaches to achieve quality localization outputs particularly for activity titles on GetYourGuide by leveraging AI-powered workflows and simulating an LLM-as-a-Judge pipeline for a curated set of titles in **German**.

Workflow

The flowchart demonstrates the workflow implementation for the case study followed by explanation for each step ([Click here](#) for higher resolution):



Data Collection

Due to inaccessibility to automated data-scraping methods through API or Web-Scraping, 10 activity titles were manually extracted from GetYourGuide website for en-US (source locale) and their machine translations for de-DE (target locale) for the purpose and scope of this *mini-project*.

Linguistic Quality Assurance (LQA)

A rubric was designed for this step to evaluate the extracted activity titles based on the linguistic quality and a final *clickability* score as the outcome from the user perspective, and the reasoning to arrive at the score.

Error Categories

The following error categories was used as a guide to identify language/linguistic quality issues in the activity titles:

Metric	Description
Accuracy	Untranslated text, mistranslations, added/missing information
Terminology	Inappropriate for travel context, inconsistency
Grammar	Syntax, capitalization, compound noun and other grammatical errors
Fluency	Awkward phrasing, translationese, unnatural word order
Formatting	All caps, invalid punctuation, invalid trailing characters

Clickability Score

Based on the identified issues, a qualitative *clickability* score for how likely the user is to click on the activity to evaluate conversion/trust impact:

Category	Description
Highly Likely	The activity title is free of language errors, following which the user is highly likely to click on the activity
Likely	The activity title contains minor language errors, but the user is still likely to click
Neutral	The language errors in the activity title do not affect the user
Not Likely	The user is unlikely to click on the activity based on the language issues in the title
Highly Unlikely	The language issues in the activity title are of a very poor quality and the user is highly unlikely to click on it

Reasoning

The last step in the evaluation was to comment on the reasoning behind deciding the clickability score. Unlike the above steps which are pre-defined and labelled, the reasoning is a free-text dimension to understand the logical thinking behind decision making - for both the Human and LLM.

Human Evaluation (Gold Data)

A human evaluation was performed on the 10 German (de-DE) activity titles based on the above LQA parameters - error categories were assigned for visible language errors and a clickability score was assigned.

In addition to the evaluation, the human evaluator also provided a suggested rewrite.

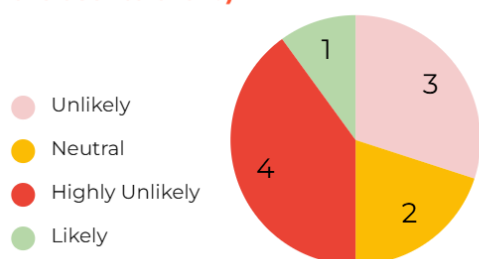
LLM-as-Judge

The same activity titles were provided as prompt-inputs along with the LQA as system prompt to the LLM (Gemini 2.5 Flash). As in the human evaluation, the LLM assigned error categories, provided a clickability score as well as suggested rewrites for the titles.

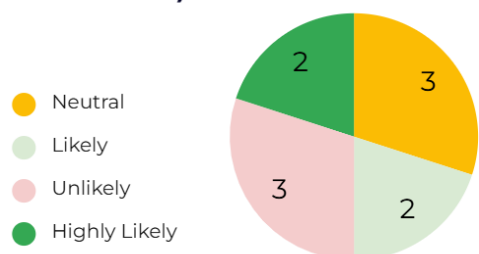
Comparison and Agreement Analysis

The Human and LLM evaluations of the 10 activity titles are compared below in terms of their agreement with each other on the same parameters namely the **clickability score** and **error categories**. It also compares the rewrite suggestions made by both, and the reasoning behind the evaluation and rewrites.

Human Clickability Score (How likely is the user to click?)

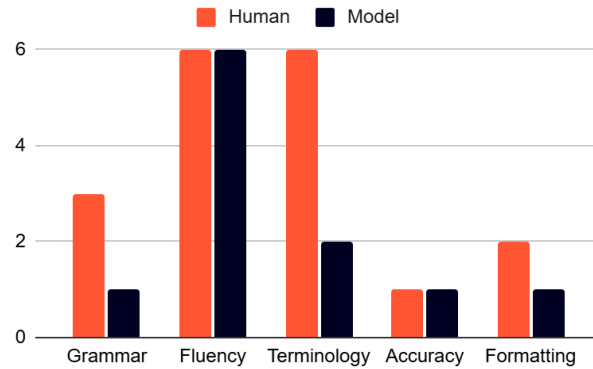


Model Clickability Score (How likely is the user to click?)



The pie charts above show a clear distinction between how human evaluation of the clickability score was more strict compared to the LLM's judgment, perhaps because the activity titles were deliberately selected for their visible issues, which introduces a slight bias in the sample.

While assigning the error categories, we see more agreement for **Fluency**, which stands out to be the most prominent issue - 6 instances out of 10. There is still disagreement on assigning **Grammar** and **Terminology** error categories.



This brings us to the next step in the workflow namely Rubric Validation and Confirmation. The **reasoning** step is introduced for these specific reasons.

One clear disagreement can be seen in the activity title, where the model clearly fails to identify an incorrect term being used.

Activity Title (de-DE): **Paris: 1-stündige Seine-Kreuzfahrt ab dem Eiffelturm**

Human Reasoning: Terminology: The term Kreuzfahrt is incorrectly used. It is used for longer cruises with cabins, etc. The correct term should be Bootsfahrt

Model Reasoning: The translation is excellent and adheres to all specified style guide rules. It correctly uses Sentence Case capitalization, forms compound nouns ('Seine-Kreuzfahrt') and adjectives ('1-stündige') appropriately, and avoids ALL CAPS. The phrasing '1-stündige Seine-Kreuzfahrt ab dem Eiffelturm' is natural, fluent, and uses standard German travel terminology, avoiding any 'translationese'. All essential information from the source is accurately conveyed. This high quality ensures user trust and a positive user experience, making conversion highly likely.

Kreuzfahrt is a terminology from tourism which is restricted to cruises of longer duration, while short one hour river cruises use the term **Bootsfahrt**. The model fails to identify this distinction.

Rubric Validation

The above analysis is a clear indication of the existing disagreement between the Human and LLM evaluation, except on the **Fluency** metric. The model also generates lengthy reasoning comments, which may affect readability.

This entails that the rubric needs a few modifications. The possible ones are listed below:

- Few-shot prompting with more examples that can be controlled through *Translation Memories* (using RAG)
- More descriptive definitions of error categories for disambiguation. This helps both human and LLM for more deterministic evaluation.
- Stricter word count for reasoning

This calls for a refinement approach until a satisfactory Rubric is attained.

Scalability

For the purposes of this demonstration, the workflow proceeds to illustrate the scaling approach; in a production setting, this step would only follow once the rubric refinement loop (above) reaches a validated state

LQA Automation

One direct scalable implementation of the preceding steps in the workflow is recycling the LQA for all other locales. This can be achieved through meta-prompting a LLM to generate the system instructions for each of them.

LLM-as-Judge (All locales)

Once the LLM-as-Judge workflow is validated for de-DE, the LQAs from above can be used to trigger it for all 36 supported locales at once.

Human-Meta Review

While the LLM-as-Judge method in itself is the ultimate scalable implementation, it requires multiple iterations of the above steps before it reaches the desired level of accuracy. This can be achieved through bringing in the human-in-the-loop: the linguistic / localization experts. The example from above for the **Kreuzfahrt** title is a demonstration of how the meta review will work.

Decision Layer

The final layer in the workflow is the decision layer, although it's a dynamic step that encompasses all the previous steps.

The eventual decisions to be taken for model deployment depend on factors listed below:

- Is the disagreement between Human and LLM-as-Judge negligible, or the LQA requires further fine-tuning?
- Are the final generated rewrite-suggestions good enough to replace the original titles?
- Is the LLM-as-Judge implementation robust enough for product wide implementation (activity description, website content, etc.)?

Appendix

1. Activity Titles (Sample Data)

Source Language (en-US)	Target Language (de-DE)
Luxor: Valley of the Kings and Queens Guided Tour with Lunch	Luxor: Tal der Könige und Königinnen Geführte Tour mit Mittagessen
Luxor: Karnak Temple and Luxor Temple Tour with Lunch	Luxor: Karnak-Tempel und Luxor-Tempel-Tour mit Mittagessen
Berlin Dungeon: Entrance Ticket	Berlin Dungeon: Ticket für den Eintritt
Berlin: Reichstag, Dome and Government District Guided Tour	Berlin: Führung durch Plenarsaal, Kuppel und Regierungsviertel
Berlin: Kinky Club Behind-the-Scenes Tour with Champagne	Berlin: Tour hinter die Kulissen eines Kinky-Clubs mit Champagner
Paris: Photoshoot by the Eiffel Tower, BEST PRICE AND VALUE	Paris: Fotoshooting am Eiffelturm, BESTER PREIS UND BESTES PREIS-LEISTUNGS-VERHÄLTNIS
Paris: Evening Lights Cruise with Legendary Bateaux-Mouches	Paris: Abendliche Lichter-Bootsfahrt mit den legendären Bateaux-Mouches
Paris: 1-Hour Seine Cruise departing from the Eiffel Tower	Paris: 1-stündige Seine-Kreuzfahrt ab dem Eiffelturm
Rome: Nighttime Tour by Open-Top Bus	Rom: Nachttour im offenen Bus
Rome: Pub Crawl & Ultimate Party Tour	Rom: Kneipentour & ultimative Party-Tour

2. GitHub Repository: <https://github.com/FootlooseNFree/GetYourGuide>

Resource Name	URL
Notebook for Workflow Implementation	https://github.com/FootlooseNFree/GetYourGuide/blob/main/GetYourGuide_CaseStudy_ColabNotebook.ipynb
System Prompt (Instruction) for the LLM	https://github.com/FootlooseNFree/GetYourGuide/blob/main/system_prompt_de.txt
Process Flowchart	https://github.com/FootlooseNFree/GetYourGuide/blob/main/GetYourGuide_CaseStudy_Flowchart.jpg
Evaluated Data (Human + LLM)	https://github.com/FootlooseNFree/GetYourGuide/blob/main/GetYourGuide_CaseStudy_EvaluatedData.csv

	etYourGuide/blob/main/GetYourGuide_CaseStudy_Gemini_Evaluated.csv
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3. Streamlit App: <https://getyourguide-case-study.streamlit.app/>

Instructions to use the app:

- Data overview can be shown/hidden (hidden by default)
- Choose the English title to view human v/s LLM comparison